Beyond Complexity Reduction: Probability Sensitivity in LLM-Based Code Generation Systems

Methodology (First Draft)

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I. Research Design

[TODO: Name the design and justify it. Which parts are experimental (manipulated surface typicality, Subproblem 2) and which are correlational or descriptive (Subproblems 1 and 3)? What are the independent, dependent and control variables? State the hypotheses for each subproblem. Relate the design to the controlled approach of McCoy et al. [1].]

II. Data Collection

A. Tasks and Materials

[TODO: Which benchmark or task source (single-function / single-file only, per your delimitations)? How many tasks, and how are they chosen? Why is that number enough (power analysis or precedent)?]

B. Constructing Matched Request Pairs

[TODO: How a "typical" and an "unusual" wording are written for each task while keeping behaviour, test suite, language, library and algorithm fixed. Who writes them, using what written guide, and how the second person checks each pair.]

C. Models and Scoring

[TODO: Which open-weight model(s) score input and output probability, and which model(s) generate code. Exact versions, decoding settings (temperature, samples per task), hardware. Total and per-token log-probability, and why both.]

D. Harness Condition (Subproblem 3)

[TODO: Which harness (e.g. Agentless), how its subprompts are captured and scored, and what the single-prompt comparison is.]

E. Procedure

[TODO: Step-by-step order of operations, so another researcher could repeat it. What is logged at each step.]

III. Data Analysis

[TODO: For each subproblem: the statistical test or model, the unit of analysis, how repeated measures on the same task are handled (e.g. paired tests or mixed-effects models), effect sizes, and correction for multiple comparisons. How the two outcome measures (passes tests; matches reference) are analysed. What result would support or refute each hypothesis.]

IV. Validity and Reliability

A. Validity of Instruments

[TODO: Construct validity: does model log-probability measure "typicality"? How Subproblem 1 checks this. Content validity of the request pairs: how you show a pair differs only in surface form.]

B. Reliability of Instruments

[TODO: Inter-rater agreement on pair equivalence (e.g. Cohen's κ, with a threshold and what happens below it). Test–retest stability of probability scores and of generated-code pass rates across seeds.]

C. Threats to Internal and External Validity

[TODO: Confounds such as prompt length, benchmark contamination, and the open-weight model standing in for closed ones. How each is controlled or measured. What the results do and do not generalise to.]

V. Ethical Considerations

[TODO: Licences of benchmarks and models, compute cost, whether any human participants (the second rater) need consent, data and code release.]

VI. Limitations

[TODO: What the design cannot show, beyond the delimitations already stated in the proposal.]

Appendix A
Use of AI Tools

[TODO: Your AI use statement for this part.]

REFERENCES

[1] R. T. McCoy, S. Yao, D. Friedman, M. D. Hardy, and T. L. Griffiths, "Embers of autoregression show how large language models are shaped by the problem they are trained to solve," *Proceedings of the National Academy of Sciences*, vol. 121, no. 41, p. e2322420121, Oct. 2024, peer-reviewed journal version of the arXiv preprint (arXiv:2309.13638, Sep. 2023); expands evaluation from 2 to 5 LLMs (adds Claude 3, Llama 3, Gemini 1.0). Citation key retained from the preprint entry so existing `\cite` calls do not need updating.